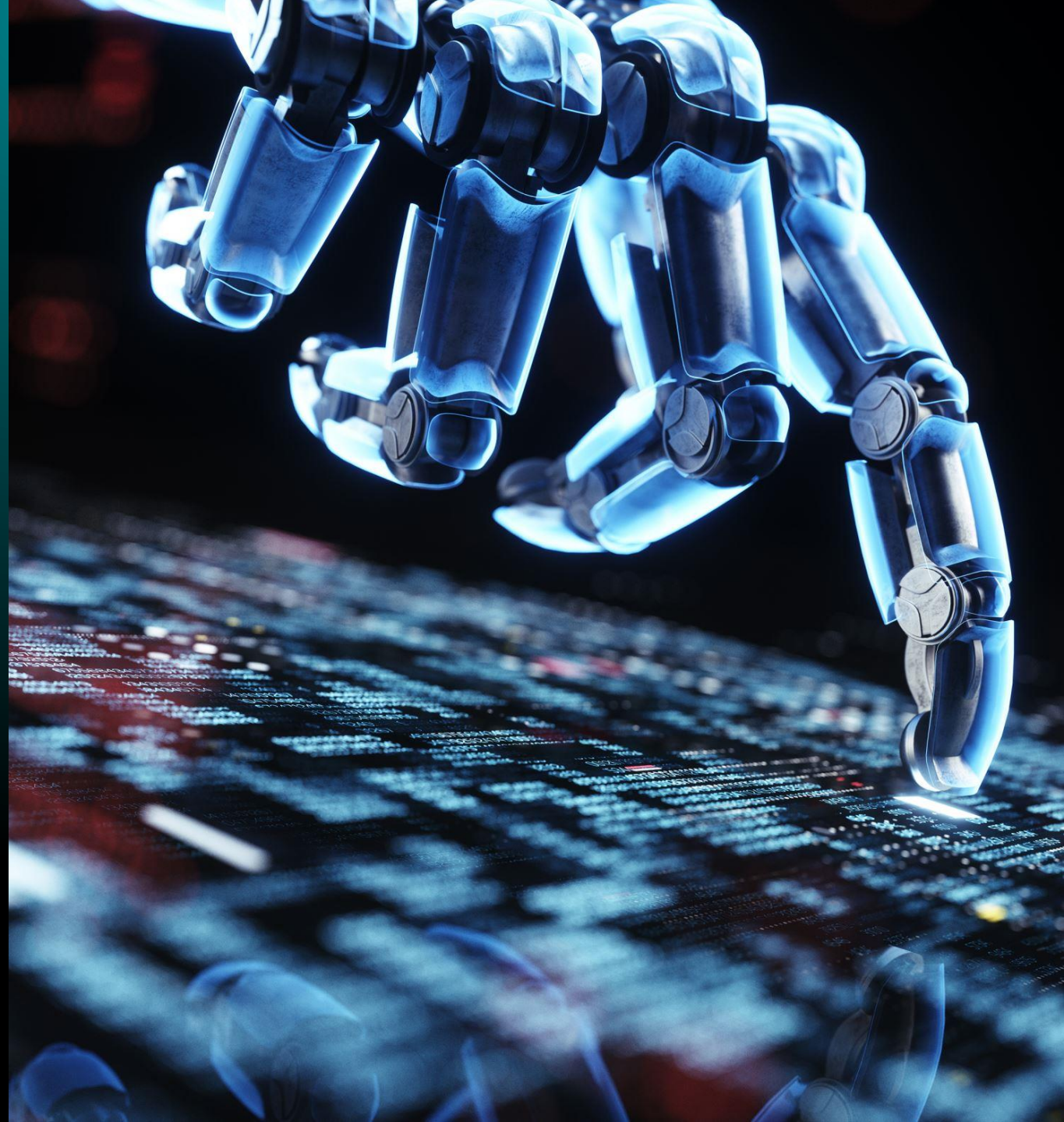
Stress Testing a Markovitz Portfolio

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Objectives

- Construct and dynamically manage a Minimum Variance Portfolio based on recent daily return data of two assets
- Estimate portfolio risk using several Value at Risk (VaR) and Expected Shortfall (ES) models
- Evaluate model performance across both stable and high-volatility periods, such as the COVID-19 pandemic and the Ukraine invasion
- Enhance risk estimation accuracy by incorporating deep learning techniques and stress-testing scenarios.

Data Collection

- The chosen assets for this project are:
  - Pfizer: a pharmaceutical company
  - Rheinmetall: a defense company
- The reason behind this choice is that these two assets were heavily influenced by market shocks caused by major events in the 2010-2023 period: the COVID-19 pandemic and the Russian invasion of Ukraine
- These assets allow to construct an adaptive portfolio capable of quickly respond to sudden changes of the market
- This is essential to make the portfolio able to behave safely under new stressed conditions, like those experienced in 2025



Portfolio Construction

- **Covariance matrix:** computed using rolling window of assets returns
- **Optimization:** minimize portfolio variance under full investment constraint
- **Result:** weight more allocated to Pfizer due to its lower volatility

Capital Allocation

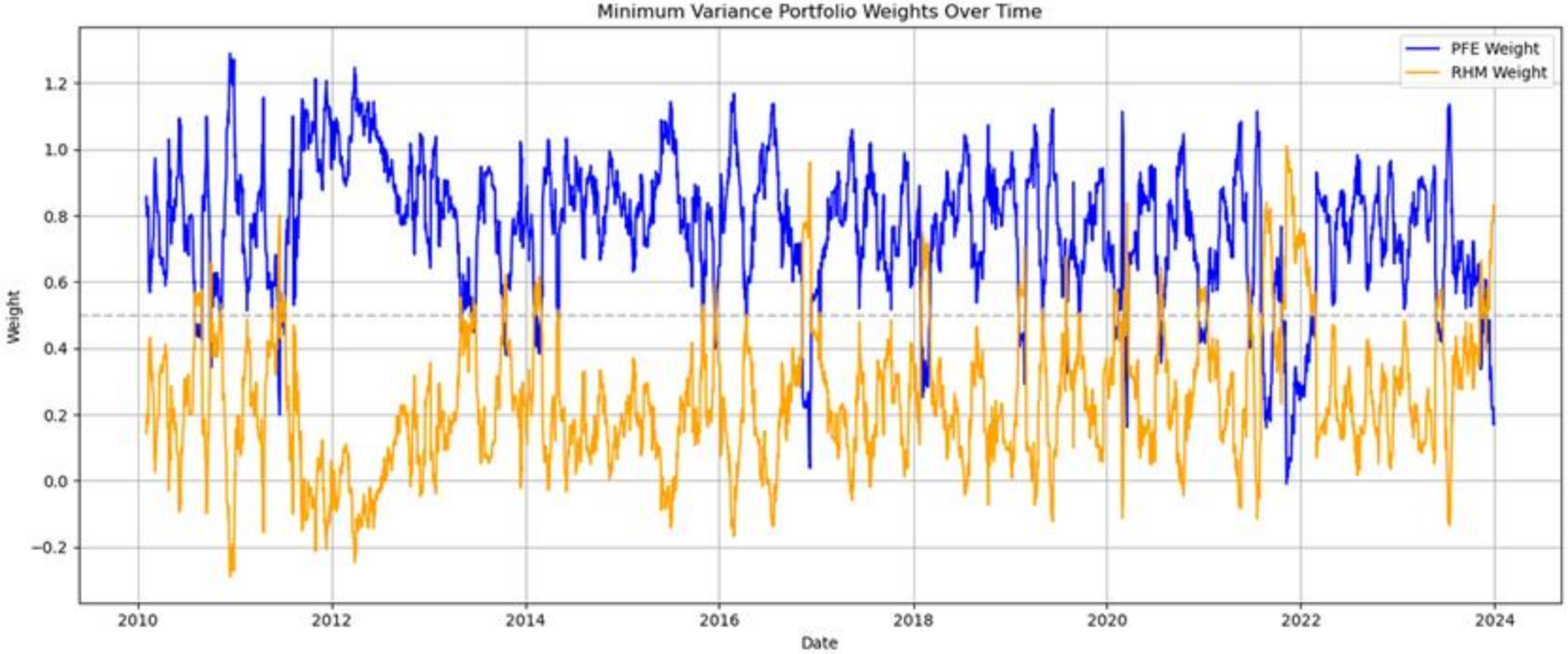
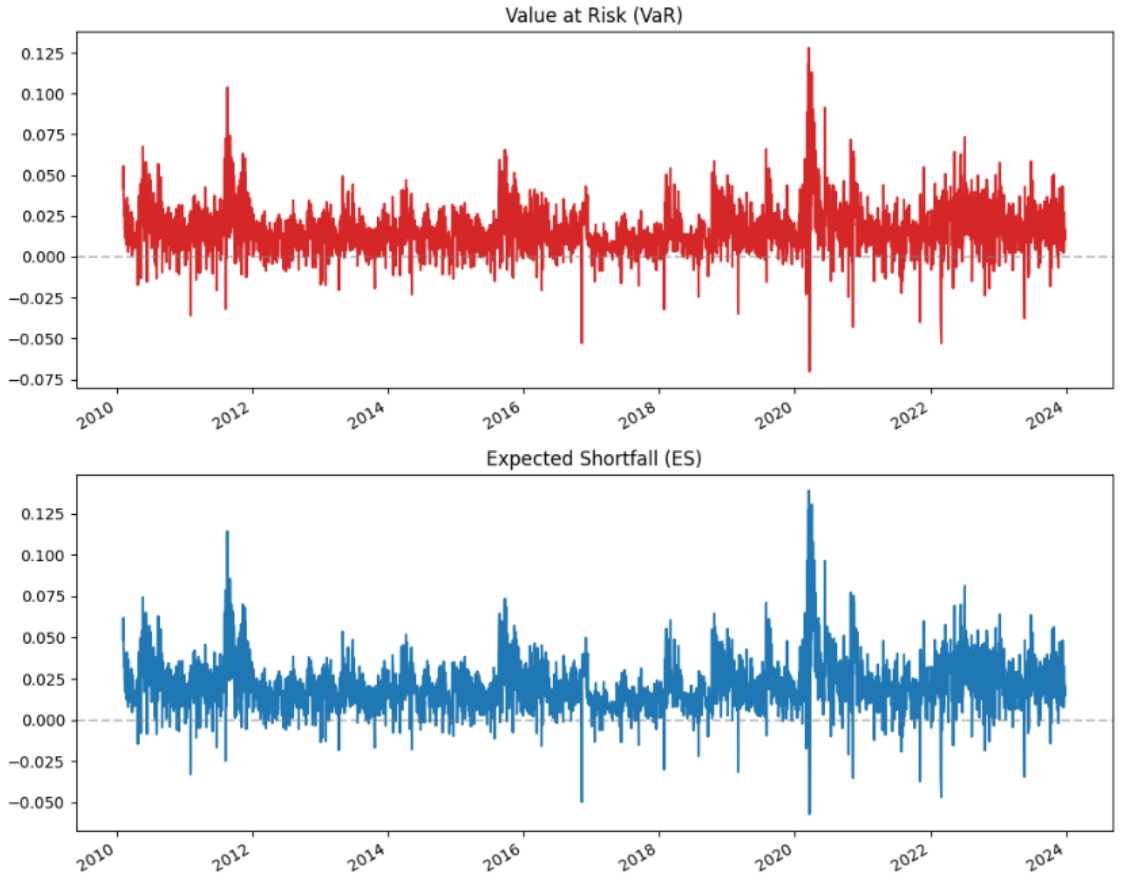
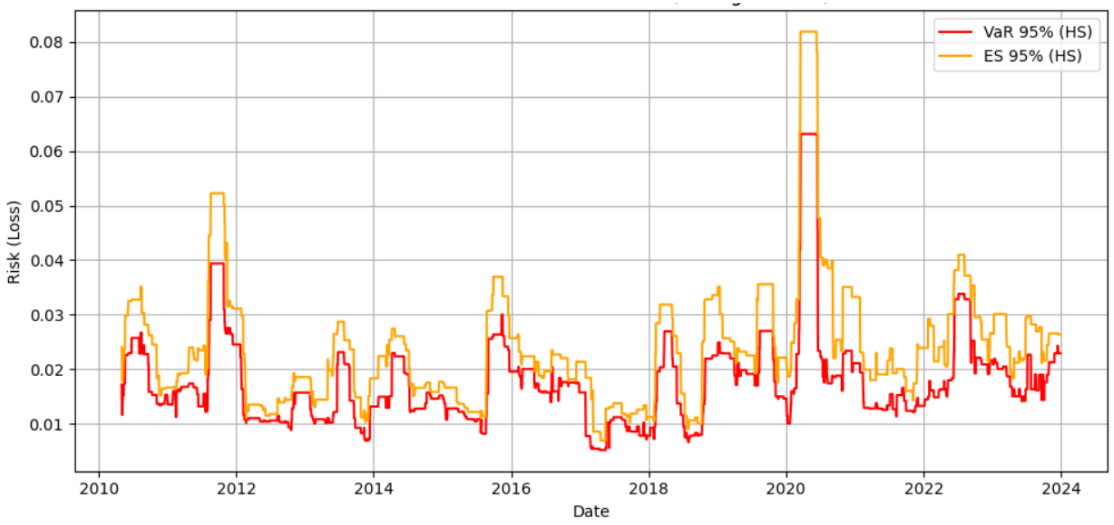


Figure 5: Daily Portfolio Weights



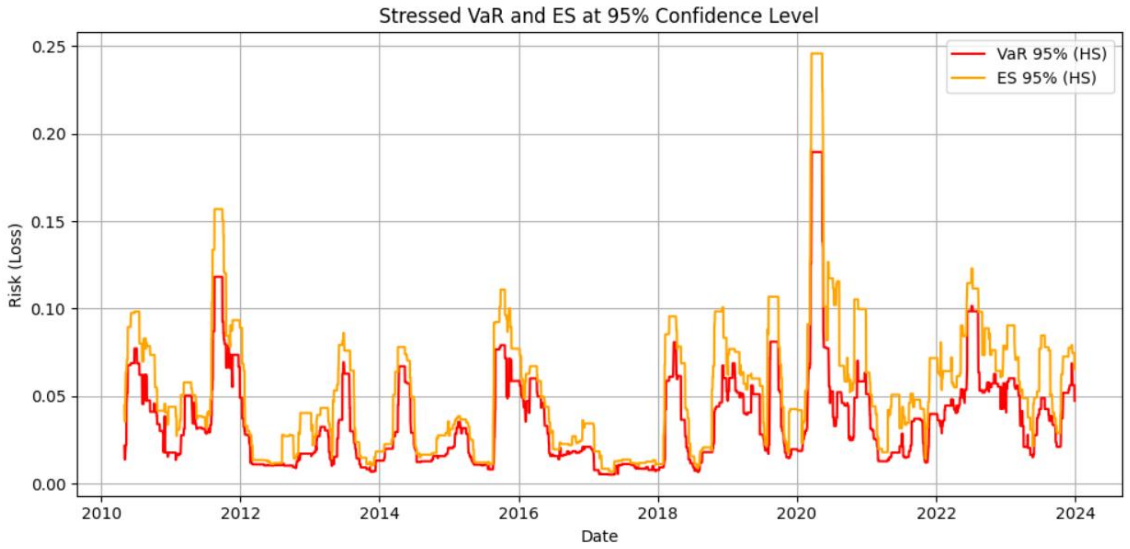
Risk Estimation-Normal Distributed Returns Assumption

- **VaR Violation Rate:** 0%
- The model is **overconservative**, consistently overestimating risk and failing to capture the actual behavior of the portfolio
- This may lead to **inefficient capital allocation**



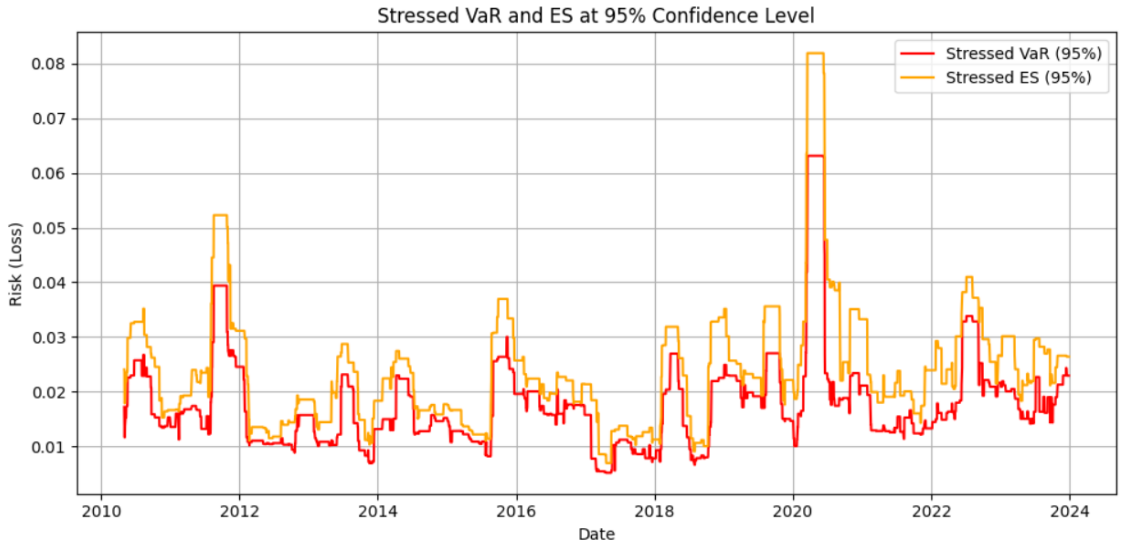
Risk Estimation- Historical Estimate

- **Non-parametric**, data-driven model
- Uses a **rolling window** over the past days to estimate future risk
- **VaR Violation Rate:** 6.14%
- On the training set it shows **strong performance** at risk forecasting
- **Main issue:** it assumes that the past is completely representative of the future



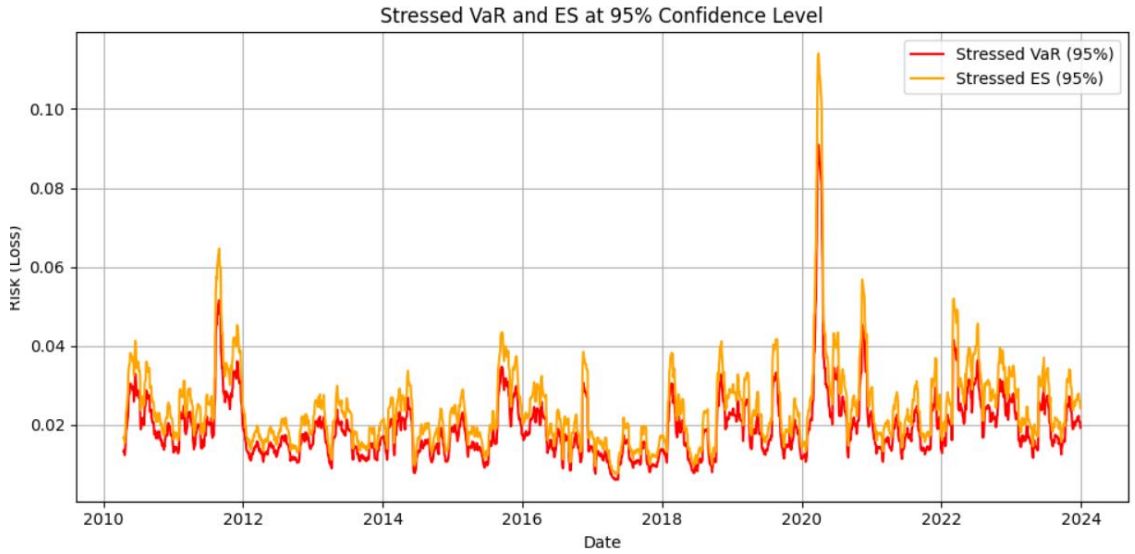
Risk Estimation– Stress Amplification Model

- **Amplifies** the **impact of high volatility days** rather than treating all past observations equally
- The model is more able to **generalize on unseen data**, preventing the risk of overfitting
- **VaR Violation Rate:** 2.78%
- The model is **overconservative**
- **Main issue:** the definition of "risk" is static



Risk Estimation– Z Score Model

- Computes **mean** and **standard deviation of volatility** for a window of past days
- Computes **z-scores** for each day of the window, estimating **how many standard deviations** a volatility observation **lies from the mean**
- It offers a **dynamic definition of "risk"**
- **VaR Violation Rate:** 6.14%
- Strong performance on the training set



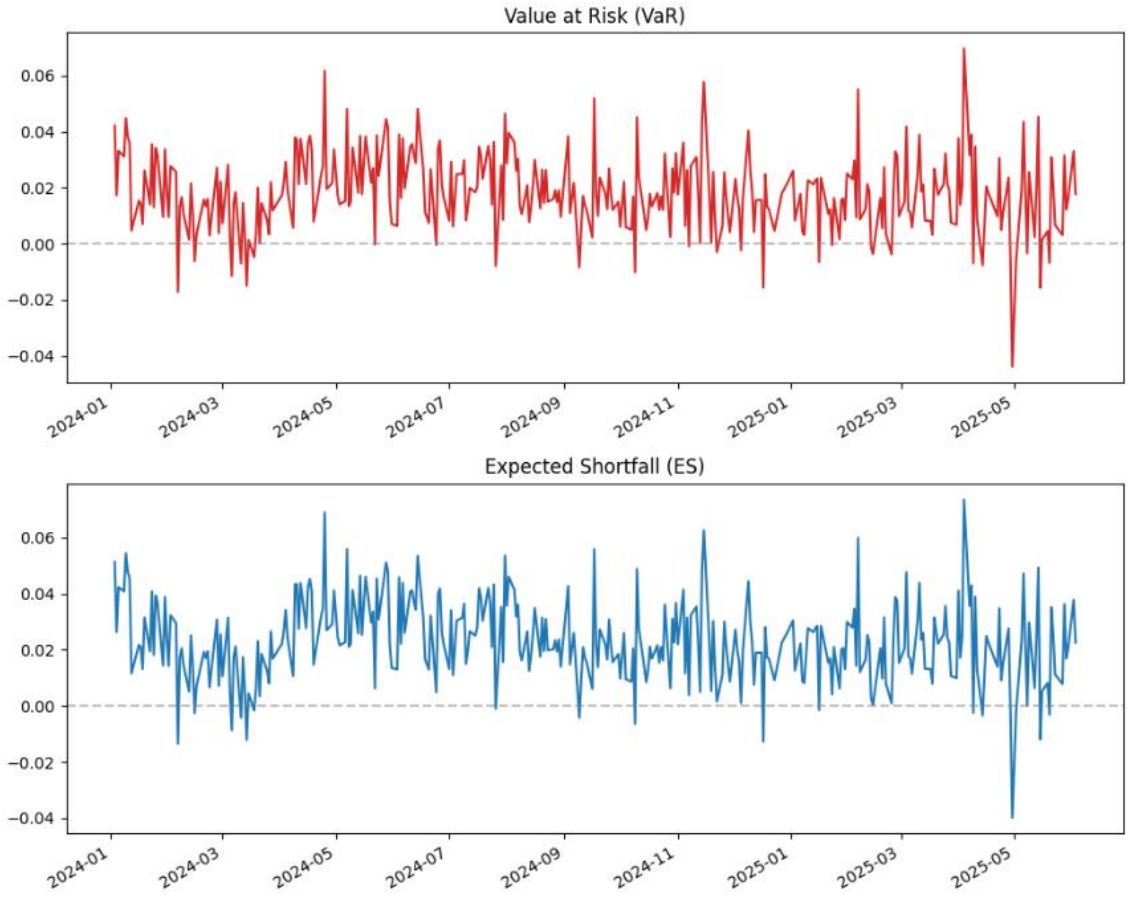
Risk Estimation- LSTM

- **LSTM** is a deep learning technique able to **capture hidden structures** among data, specifically designed for sequential data
- **VaR Violation Rate:** 4.8%
- **Performance** on training set is the **best so far**, showing LSTM's ability to forecast future events based on recent observations



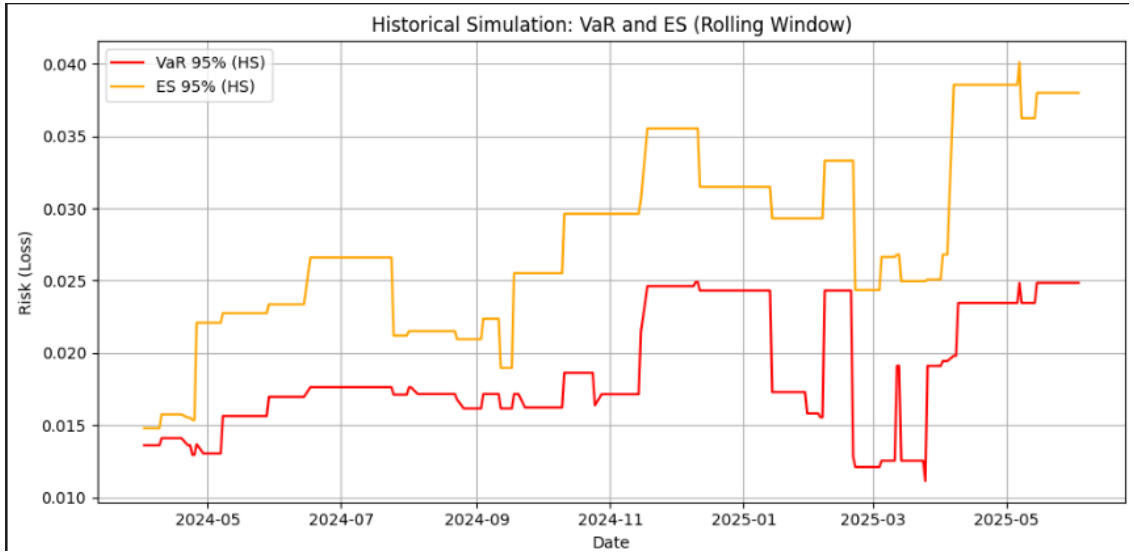
Model Evaluation

- To evaluate the performance of the models the 2024-2025 period was considered
- Daily covariance matrix and portfolio weights were computed as previously explained
- The plot shows the daily value of the portfolio in the test period



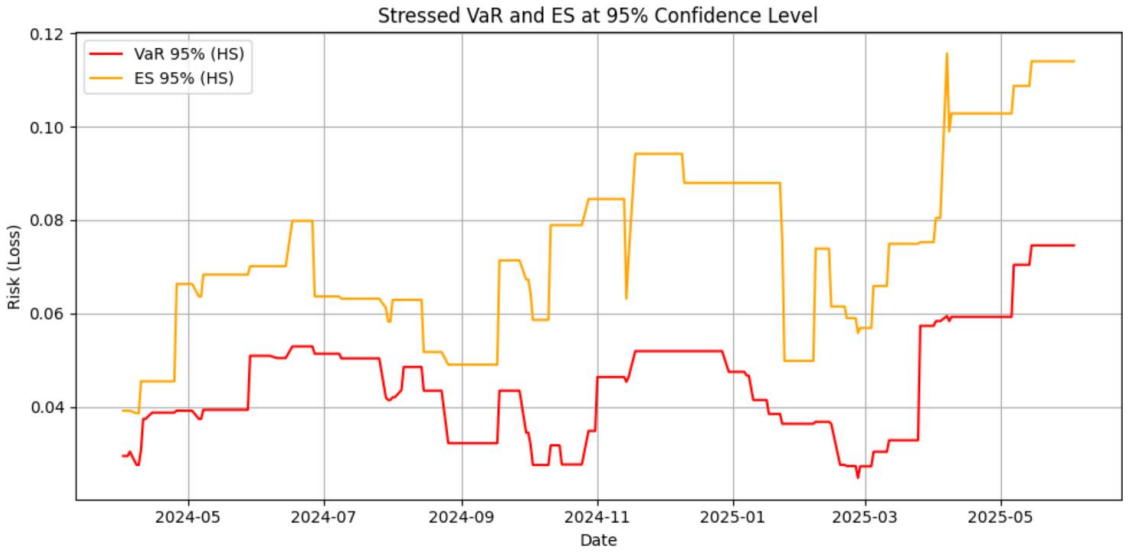
Model Evaluation- Normal Distributed Returns Assumption

- **VaR Violation Rate:** 0%
- The model is **unable to capture the real behavior** of the portfolio, as it happened on the training set
- The model is clearly **overconservative**



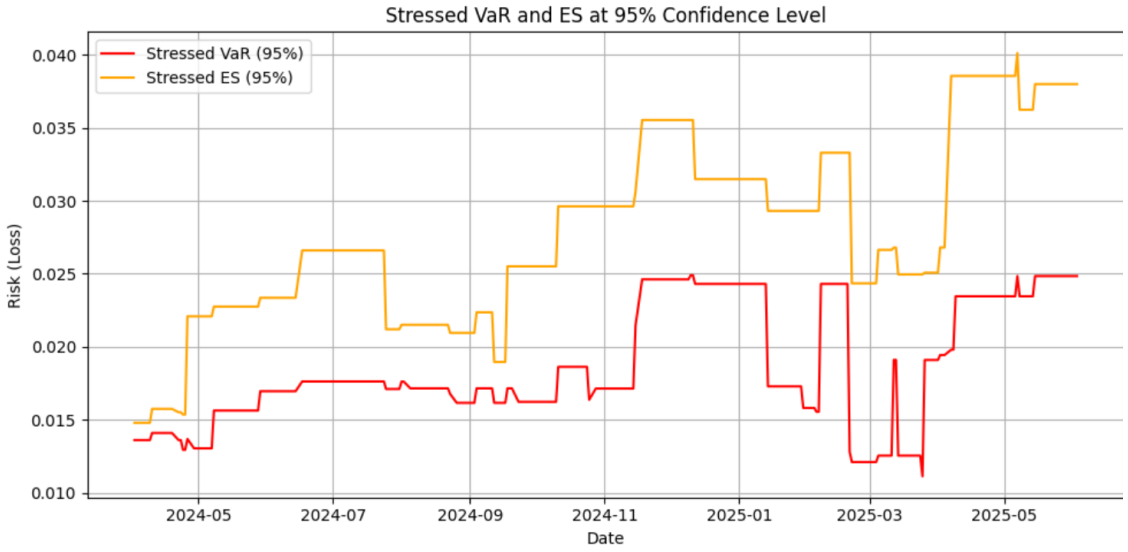
Model Evaluation- Historical Estimate

- **VaR Violation Rate:** 7.29%
- The model **slightly underestimates the risk**, but the Violation rate is close to the target



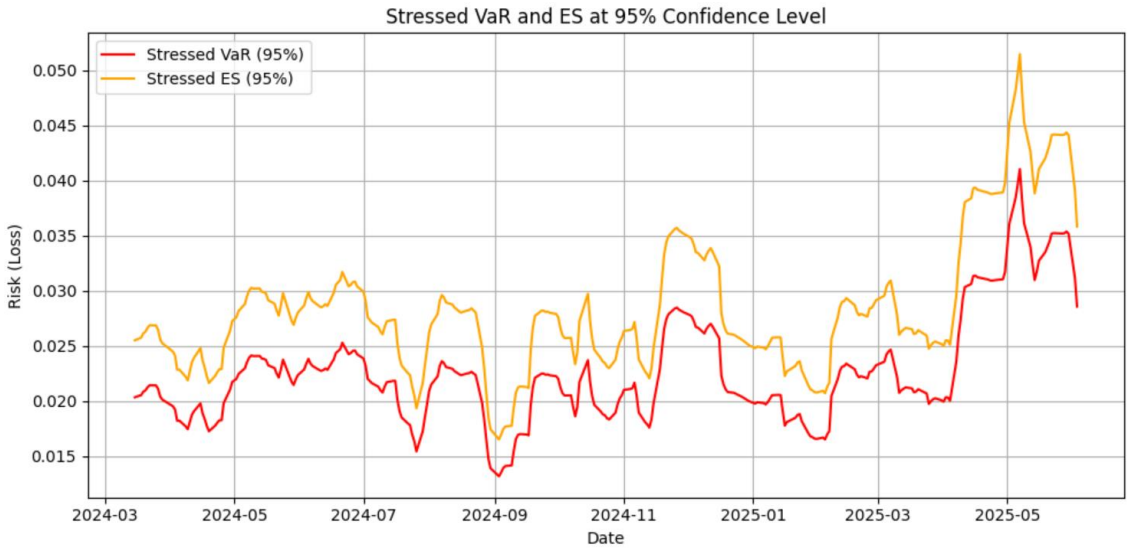
Model Evaluation-Stress Amplification

- **VaR Violation Rate:** 0.69%
- The model clearly **suffers of overfitting**, showing poor performance on the test set
- The model is **too conservative,** leading to poor capital allocation



Model Evaluation-Z Score Model

- **VaR Violation Rate:** 7.29%
- As the Historical Estimate-based model, the model **slightly underestimates risk**, exposing to potential losses
- Overall, the performance is good, as the Violation Rate is close to the target, showing that the dynamic definition of risk **adapts well to new dynamic scenarios**



Model Evaluation- LSTM

- **VaR Violation Rate:** 3.36%
- Very close to the target, the model shows a very good ability to adapt to new stressed conditions, without excessively overestimating risk
- The LSTM model confirms the best at predicting future events based on hidden patterns among data

	Model	VaR Violation Rate (%)	Actual ES	Predicted ES
0	Normal Distribution	0.00	NaN	0.023002
1	Historical	7.29	0.0266	0.027533
2	Stress Amplification	0.69	0.0338	0.073678
3	Stress Amplification (Z-Score)	7.29	0.0266	0.027533
4	LSTM	3.36	0.0329	0.028303

Risk Estimation–Model Comparison

- Returns **normally distributed** assumption-based model is far **too conservative**
- **Historical estimate** model and **Z-Score model** share the same performance, slightly **underestimating risk**
- **LSTM** model shows the **best performance**, avoiding unnecessary risk (at the cost of being a bit overconservative)

Conclusion

- In this project several risk-estimation models were tested, calibrated, and evaluated
- Static models such as **Normal distribution assumption-based model** and **Stress Amplification model** appear to be **overconservative**, failing to capture the real behavior of the market
- More dynamic models such as **Z-Score model** are capable of **estimating risk** quite **efficiently**
- **Deep learning techniques** such as **LSTM** emerge as the **best models for risk forecasting**, as they can learn hidden patterns among the data other models will miss

Thanks for your attention!